

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Database Management Systems

COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO1: *Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships. (BL1, BL2)*

Activity Tool : Short Answer Text(High)

Assessment Tool Weightage: 30%

Dr

QUESTIONS		Total Marks: 50	
Q.No	Question	Bloom's Level	Marks
1	UPS Product Tracking Information System UPS prides itself on having up-to-date information on the processing and current location of each shipped item. To do this, UPS relies on a company-wide information system. Shipped items are the heart of the UPS product tracking information system. Shipped items can be characterized by item number (unique), weight, dimensions, insurance amount, destination, and final delivery date. Shipped items are received into the UPS system at a single retail center. Retail centers are characterized by their type, uniqueID, and address. Shipped items make their way to their destination via one or more standard UPS transportation events (i.e., flights, truck deliveries). These transportation events are characterized by a unique scheduleNumber, a type	BL1 – Remember	5

	(e.g, flight, truck), and a deliveryRoute. Create an Entity Relationship diagram that captures this information about the UPS system.		
2	Course Management System Consider a university database for the scheduling of classrooms for final exams. This database could be modeled as the single entity set exam, with attributes course-name, section-number, room-number, and time. Alternatively, one or more additional entity sets could be defined, along with relationship sets to replace some of the attributes of the exam entity set, as • course with attributes name, department, and c-number • section with attributes s-number and enrollment, and dependent as a weak entity set on course • room with attributes r-number, capacity, and building Show an E-R diagram illustrating the use of all three additional entity sets listed.	BL2 – Understand	5
3	University Registrar's Office Management System A university registrar's office maintains data about the following entities: (a) courses, including number, title, credits, syllabus, and prerequisites; (b) course offerings, including course number, year, semester, section number, instructor(s), timings, and classroom; (c) students, including student-id, name, and program; (d) instructors, including identification number, name, department, and title. (e) Further, the enrollment of students in courses and grades awarded to students in each course they are enrolled for must be appropriately modeled. Construct an E-R diagram for the registrar's office.	BL2 – Understand	5

4	Hospital Management System Construct an E-R diagram for a hospital with a set of patients and a set of medical doctors. Associate with each patient a log of the various tests and examinations conducted. Construct appropriate tables for the above ER Diagram : Patient(SS#, name, insurance) Physician (name, specialization) Test-log(SS#, test-name, date, time) Doctor-patient (physician-name, SS#) Patient-history(SS#, test-name, date)	BL2 – Understand	5
5	Company Project Management System Construct an ER Diagram for Company having following details : Company organized into DEPARTMENT. Each department has unique name and a particular employee who manages the department. Start date for the manager is recorded. Department may have several locations. □ A department controls a number of PROJECT. Projects have a unique name, number and a single location. □ Company's EMPLOYEE name, ssno, address, salary, sex and birth date are recorded. An employee is assigned to one department, but may work for several projects (not necessarily controlled by dept). Number of hours/week an employee works on each project is recorded; The immediate supervisor for the employee. □ Employee's DEPENDENT are tracked for health insurance purposes (dependentname, birthdate, relationship to employee).	BL2 – Understand	5
6	Explain schema and instance with examples. Distinguish between logical schema, physical schema, and view schema.	BL1 – Remember	5
7	Identify the entities, attributes, and relationships for a Bank Management System and construct its ER diagram.	BL2 – Understand	5

8	Explain the Extended ER (EER) model. Describe the concepts of generalization, specialization, and aggregation.	BL2 – Understand	5
9	Design an EER diagram for a Hospital Management System incorporating inheritance and weak entities.	BL2 – Understand	5
10	Explain the role of a DBA (Database Administrator) and the responsibilities associated with the position.	BL1 – Remember	5

Code of Conduct:

I KESHAVARAM L/192511253 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: KESHAVARAM L

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	30%	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Linkages	25%	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Organization	20%	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application

Integration of Literature	15%	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity	10%	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

Q1. UPS Product Tracking Information System (ER Diagram)

Entities

Shipped_Item

- Item_No (PK)
- Weight
- Dimensions
- Insurance_Amount
- Destination
- Final_Delivery_Date

Retail_Center

- Center_ID (PK)
- Type
- Address

Transportation_Event

- Schedule_No (PK)
- Type (Flight/Truck)
- Delivery_Route

Relationships

- Retail_Center **receives** Shipped_Item (1:N)
- Shipped_Item **passes through** Transportation_Event (M:N)

ER Diagram (Text)

Retail_Center

Center_ID (PK)

Type

Address

|

| Receives (1:N)

|

Shipped_Item

Item_No (PK)

Weight

Dimensions

Insurance

Destination

Delivery_Date
 |
 | Travels Through (M:N)
 |
 Transportation_Event

 Schedule_No (PK)
 Type
 Delivery_Route

Q2. Course Management System (ER Diagram)

Entities

Course

- Course_Name (PK)
- Department
- Course_No

Section (Weak Entity)

- Section_No
- Enrollment

Room

- Room_No (PK)
- Capacity
- Building

Exam

- Time

Relationships

- Course **has** Section (1:N)
- Section **scheduled in** Room
- Section **has** Exam

Course

 Course_No(PK)
 Name
 Department

|
 | Has
 |

Section (Weak)

 Section_No
 Enrollment

|
 | Scheduled In
 |

Room

Room_No(PK)

Capacity

Building

Section

|

| Has

|

Exam

Time

Q3. University Registrar's Office ER Diagram

Entities

Course

- Course_No (PK)
- Title
- Credits
- Syllabus
- Prerequisites

Course_Offering

- Year
- Semester
- Section_No
- Timing
- Classroom

Student

- Student_ID (PK)
- Name
- Program

Instructor

- Instructor_ID (PK)
- Name
- Department
- Title

Relationships

- Course → Course Offering (1:N)
- Instructor teaches Course Offering
- Student enrolls in Course Offering
- Enrollment contains Grade

Course ----< Course_Offering >---- Instructor

|

|
Enrollment
|
Student
(Grade)

Q4. Hospital Management System

ER Diagram

Entities

Patient

- SS# (PK)
- Name
- Insurance

Physician

- Name (PK)
- Specialization

Test_Log

- SS#
- Test_Name
- Date
- Time

Patient_History

- SS#
- Test_Name
- Date

Relationship

Physician treats Patient (M:N)

Tables

PATIENT

SS# (PK)

Name

Insurance

PHYSICIAN

Name (PK)

Specialization

TEST_LOG

SS#

Test_Name

Date

Time

DOCTOR_PATIENT

Physician_Name

SS#

PATIENT_HISTORY

SS#

Test_Name

Date

Q5. Company Project Management System

Entities

Department

- Dept_Name (PK)
- Manager
- Start_Date
- Location

Project

- Project_No (PK)
- Project_Name
- Location

Employee

- SSN (PK)
- Name
- Address
- Salary
- Sex
- Birth_Date

Dependent

- Dependent_Name
- Birth_Date
- Relationship

Relationships

- Department controls Project
- Department employs Employee
- Employee works on Project
- Employee supervises Employee
- Employee has Dependents

Q6. Schema and Instance

Schema

Schema is the logical structure or blueprint of a database.

Example

STUDENT(RegNo, Name, Department)

Instance

Instance is the actual data stored in the database at a particular time.

Example

RegNo Name Department

101 Rahul CSE

102 Anita AI

Types of Schema

Logical Schema

- Defines tables, attributes and relationships.
- Independent of storage.

Physical Schema

- Defines how data is stored physically.
- Includes files and indexes.

View Schema

- Defines user-specific views.
 - Users see only required data.
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Q7. Bank Management System

Entities

Customer

- Customer_ID
- Name
- Address
- Phone

Account

- Account_No
- Balance
- Type

Branch

- Branch_ID
- Branch_Name

Loan

- Loan_No
- Amount

Relationships

- Customer owns Account
 - Branch maintains Account
 - Customer takes Loan
 - Branch provides Loan
-

Q8. Extended ER (EER) Model

EER is an extension of the ER model that supports inheritance and advanced relationships.

Generalization

Combines similar lower-level entities into a higher-level entity.

Example:

Car

Bike

|

Vehicle

Specialization

Divides a higher-level entity into subclasses.

Example:

Employee

Faculty Staff

Aggregation

Treats a relationship as a higher-level entity.

Example:

Employee ---- Works_On ---- Project

(Works_On) becomes an aggregate entity.

Q9. Hospital Management System (EER)

Superclass

Person

- Person_ID
- Name
- Address

Subclasses

Patient

- Insurance_No

Doctor

- Specialization

Weak Entity

Test_Record

- Test_Name
- Date
- Time

Relationships

- Doctor treats Patient
- Patient owns Test_Record

Person

/ \

Patient Doctor

|

| Owns

|
Test_Record (Weak)

Doctor

|
Treats

|
Patient

Q10. Role of DBA (Database Administrator)

Responsibilities

1. Database installation and configuration.
2. User account and security management.
3. Backup and recovery.
4. Performance monitoring and tuning.
5. Database maintenance.
6. Data integrity and consistency.
7. Storage management.
8. Disaster recovery planning.
9. Access control and authorization.
10. Troubleshooting database issues.

Functions of DBA

- Create databases.
- Manage users and permissions.
- Monitor database performance.
- Perform backups and recovery.
- Ensure database security.
- Maintain availability and reliability.